# PERIODIC GAUSSIAN LATTICE LIMITS IN TWO AND FOUR DIMENSIONS: A COMMON PROOF MECHANISM AND ITS LEAN REALIZATION

YJ ZHENG

Abstract. We connect two Lean developments of fixed-volume periodic Gaussian lattice limits: a two-dimensional limit on a Fourier realization of $H^{-3}$ and a four-dimensional raw atomic-lift limit on a Fourier realization of $H^{-4}$. Their common mechanism is a spacing-uniform bound on Fourier-mode moments, a compact inclusion between negative Sobolev carriers, convergence of finite Gaussian marginals, and uniqueness from countably many Fourier coordinates. We give a dimension-independent paper proof for raw lifts in $H^{-s}$ whenever $s > d/2$, and distinguish it from the two formally verified instantiations. The comparison makes explicit the site-covariance normalization, lattice aliases, the zero and Nyquist modes, real support, and the difference between stored weighted coordinates and physical coefficients. It also explains why the raw atomic approximation requires a rougher space than the limiting free field itself. The accompanying sources include typed theorem contracts, source and dependency hashes, axiom transcripts, symbolic scale checks, and an audit of the compiled article.

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## 1. Introduction and the connection between the proofs

The continuum limit of a lattice Gaussian field is conceptually elementary: its finite-dimensional laws are Gaussian and its covariance has a Fourier symbol. A proof on a space of distributions nevertheless needs more than pointwise convergence of that symbol. It must place every lattice

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law on a common measurable carrier, prove tightness there, and show that Fourier data determine every possible limit. These requirements are especially visible when the embedding is a sum of Dirac masses rather than an interpolation.

This article compares two completed instances of that problem. In two spatial dimensions the formal construction controls an $H^{-2}$ energy and proves weak convergence on a concrete $H^{-3}$ carrier. In four dimensions it controls an $H^{-3}$ energy and proves convergence on an $H^{-4}$ carrier to an explicitly constructed Gaussian-series law. These are related proofs, not a claim that changing a dimension parameter in the first Lean theorem produces the second. The four-dimensional summability, carrier maps and real-coordinate reconstruction have their own proofs.

Our main mathematical synthesis is the following statement. We use a Fourier Sobolev realization defined precisely in Section 3.

**Theorem 1.1** (Common paper theorem)**.** *Let $d \geq 1$ be an integer, $L > 0$, and $m > 0$. Put $a = L/N$ for positive integers $N$. On $(\mathbb{Z}/N\mathbb{Z})^d$ let $\phi_N$ be the centered real Gaussian with site covariance $a^{-d}(m^2 - \Delta_a)^{-1}$, and define*

$$X_N = a^d \sum_x \phi_N(x)\delta_{[ax]}. \tag{1}$$

*For every $s > d/2$, the laws of $X_N$ converge weakly on $\mathcal{H}^{-s}(\mathbb{T}_L^d)$ to the unique centered real periodic Gaussian field with covariance*

$$Q(f,g) = L^{-d} \sum_{k \in \mathbb{Z}^d} \frac{\overline{Ff(k)}Fg(k)}{m^2 + (2\pi/L)^2|k|_2^2}, \qquad Ff(k) = \int_{\mathbb{T}_L^d} f(x)e^{-2\pi ik\cdot x/L}\,\mathrm{d}x \tag{2}$$

*for real smooth periodic $f, g$.*

*Evidence boundary.* Theorem 1.1 is a paper theorem proved below for arbitrary $d$ and admissible $s$. The public Lean library verifies the fixed choices $(d, r, s) = (2, 2, 3)$ and $(4, 3, 4)$, where $r$ denotes the stronger moment space. It does not contain an arbitrary-dimensional formalization of this theorem.

The useful connection is the chain

$$\begin{gathered}\text{finite lattice covariance and raw Fourier coefficients}\\ \Downarrow\\ \text{uniform stronger-space moment and compact weaker-space balls}\\ \Downarrow\\ \text{tightness, finite Gaussian marginal limits, and cylinder uniqueness}\\ \Downarrow\\ \text{weak convergence of the full sequence.}\end{gathered} \tag{3}$$

Dimension enters the summability threshold and the compact carrier estimates. The final probability argument has the same structure in both cases.

1.1. **Context and scope.** Fourier and nearest-neighbour descriptions of Gaussian free fields are classical; Sheffield's introduction provides background [1]. Discrete-to-continuous limits on tori in arbitrary dimension also occur for grounded polyharmonic fields and associated random measures [2]. Those models have different covariance powers and grounding conventions. Our subject is the positive-mass resolvent and the unmodified atomic lift. We make no priority claim for its classical convergence mechanism. The contribution here is a comparison of two concrete formal proof chains, a self-contained normalization and tightness argument, and reproducible evidence that identifies precisely which statements the Lean kernel checks.

All volumes and masses in the article remain fixed and positive. Neither an infinite-volume limit nor an interacting four-dimensional field follows from this result. An interacting two-dimensional

construction requires additional renormalization and uniform integrability arguments. Osterwalder–Schrader reconstruction is outside the present claims.

## 2. The lattice model and its exact Fourier data

Write $\Lambda_N = (\mathbb{Z}/N\mathbb{Z})^d$ and choose any complete set of residues for its sites. For a real configuration $u$, define

$$(\Delta_a u)(x) = a^{-2}\sum_{j=1}^{d}\big(u(x+e_j)+u(x-e_j)-2u(x)\big).$$

With the counting inner product on configurations, translation and reindexing give

$$\sum_x u(x)(-\Delta_a u)(x) = a^{-2}\sum_{j,x}\big(u(x+e_j)-u(x)\big)^2 \geq 0.$$

Consequently $Q_N = m^2 - \Delta_a$ is symmetric positive definite. Its inverse $G_N = Q_N^{-1}$ defines the site covariance $a^{-d}G_N$. The same law is the normalized density proportional to

$$\exp\left(-\frac{a^d}{2}\sum_x u(x)(Q_N u)(x)\right)\prod_x \mathrm{d}u(x).$$

This specifies both the physical covariance normalization and the finite probability law, without a continuum limit assumption.

For $p \in \Lambda_N$, characters diagonalize $Q_N$, with eigenvalues

$$\lambda_N(p) = m^2 + 4a^{-2}\sum_{j=1}^{d}\sin^2(\pi p_j/N). \tag{4}$$

The continuum symbol is $\lambda(k) = m^2 + (2\pi/L)^2\sum_j k_j^2$. We use unnormalized physical coefficients

$$\widehat{X}_N(k) = a^d\sum_{x\in\Lambda_N}\phi_N(x)e^{-2\pi ik\cdot x/N}. \tag{5}$$

For a distribution $u$ this convention means $\widehat{u}(k) = \langle u, e^{-2\pi ik\cdot(\cdot)/L}\rangle$. For a smooth real test, $\langle u, f\rangle = L^{-d}\sum_k \widehat{u}(k)\overline{Ff(k)}$ whenever this pairing is defined.

**Lemma 2.1** (Aliases and the physical volume factor)**.** *Let $p_N(k)$ denote the residue class of $k$. Then*

$$\mathbb{E}[\widehat{X}_N(k)\overline{\widehat{X}_N(\ell)}] = \frac{L^d\,\mathbf{1}_{p_N(k)=p_N(\ell)}}{\lambda_N(p_N(k))}, \tag{6}$$

$$\mathbb{E}[\widehat{X}_N(k)\widehat{X}_N(\ell)] = \frac{L^d\,\mathbf{1}_{p_N(k)=-p_N(\ell)}}{\lambda_N(p_N(k))}. \tag{7}$$

*In particular $\mathbb{E}|\widehat{X}_N(k)|^2 \leq L^d/m^2$ uniformly in $N, k$.*

*Proof.* The inverse kernel is

$$G_N(x,y) = N^{-d}\sum_{p\in\Lambda_N}\lambda_N(p)^{-1}e^{2\pi ip\cdot(x-y)/N}.$$

Insert this and the factor $a^{-d}$ into the covariance of the sums in (5). Each character sum equals $N^d$ when its frequency vanishes modulo $N$ and zero otherwise. The overall factor is $a^{2d}a^{-d}N^{-d}N^{2d} = (aN)^d = L^d$. Conjugating a coefficient reverses its frequency, proving both identities. The bound follows from $\lambda_N \geq m^2$. □

The zero frequency is a real variable with variance $L^d/m^2$. At even $N$, self-opposite residues also occur: a residue is self-opposite precisely when each coordinate is 0 or $N/2$. Its sine coefficient vanishes. Thus nonzero continuum labels are not all independent at a fixed lattice size, and treating every one of them as a separate complex Gaussian is incorrect. Formulas (6)–(7) retain these cases automatically. For any fixed finite set of integer labels, all unintended sum and difference collisions disappear for sufficiently large $N$.

**Lemma 2.2** (Fixed-mode error with physical scale)**.** *For every $k \in \mathbb{Z}^d$ and every $N \geq 1$,*

$$0 \leq \lambda(k) - \lambda_N(p_N(k)) \leq \frac{4\pi^4 a^2}{3L^4} \sum_j k_j^4, \tag{8}$$

$$0 \leq \frac{L^d}{\lambda_N(p_N(k))} - \frac{L^d}{\lambda(k)} \leq \frac{4\pi^4 a^2 L^{d-4}}{3m^4} \sum_j k_j^4. \tag{9}$$

*Proof.* For real $t$, $|\sin t| \leq |t|$ and $|t - \sin t| \leq |t|^3/6$, so $0 \leq t^2 - \sin^2 t \leq t^4/3$. The squared sine is independent of the residue representative. Set $t = \pi k_j/N = \pi a k_j/L$ and multiply the bound by $4a^{-2}$. Summing proves (8). The difference of reciprocals has numerator $\lambda - \lambda_N$ and denominator at least $m^4$; multiplication by $L^d$ proves (9). □

*Remark* 2.3 (A scale correction recorded by the article)*.* An earlier accompanying proof guide omitted $L^{-4}$ in the dispersion bound and consequently retained a spurious $L^4$ in the four-dimensional variance-error bound. Equations (8)–(9) correct those paper constants. In dimension four the factor $L^{d-4}$ is one. The final Lean weak-convergence theorem uses its own proved moment and covariance convergence results and is unaffected by this guide error. The article's audit includes a small-volume counterexample to the omitted factor, rather than checking only $L = 1$ where the defect is invisible.

*Evidence boundary.* Lemma 2.1 is the exact paper Fourier calculation. The formal four-dimensional proof uses the weaker bound $\mathbb{E}|\widehat{X}_N(k)|^2 \leq 2L^4/m^2$, obtained from separate real cosine/sine moment bounds. The stronger exact complex formula is not asserted to be a separately packaged all-$N$ Lean theorem.

## 3. Fourier carriers, atoms, and the common compactness argument

Let $R(k) = \max_{1\leq j\leq d}|k_j|$, and put $w_s(k) = (1 + R(k))^{-2s}$. We use the real Hilbert space

$$\mathcal{H}^{-s} = \left\{ z : \mathbb{Z}^d \longrightarrow \mathbb{C} : z(-k) = \overline{z(k)}, \quad \|z\|_{-s}^2 := \sum_k w_s(k)|z(k)|^2 < \infty \right\}.$$

It is the closed real part of a complex weighted sequence Hilbert space. Writing $x(k) = \sqrt{w_s(k)}z(k)$ gives an isometry to a closed real subspace of $\ell^2(\mathbb{Z}^d;\mathbb{C})$. This last space is the concrete form used in Lean: its stored coordinate is $x(k)$, not $z(k)$. A physical evaluation is $z(k) = x(k)/\sqrt{w_s(k)}$, a continuous map for each fixed $k$.

These sequences are periodic distributions. Smooth tests have rapidly decaying Fourier coefficients; Cauchy–Schwarz makes $L^{-d}\sum_k z(k)\overline{Ff(k)}$ a continuous distributional pairing. Conversely the pairing recovers all coefficients. Equivalence with the usual Fourier Sobolev topology follows from

$$R(k)^2 \leq |k|_2^2 \leq dR(k)^2, \qquad (1 + R(k))^2 \leq 2(1 + |k|_2^2) \leq 2d(1 + R(k))^2.$$

Fixed $L > 0$ also makes the Bessel weight involving $(2\pi/L)^2|k|_2^2$ equivalent. This is an analytic identification of topologies. It is not a claim that the published Lean types come with a separately proved isomorphism to a pre-existing distribution-library Sobolev type.

**Lemma 3.1** (Summability and compact inclusion)**.** *If $r > d/2$, then $S_{d,r} := \sum_k w_r(k) < \infty$. If $s > r$, every closed bounded ball of $\mathcal{H}^{-r}$ has compact closure in $\mathcal{H}^{-s}$.*

*Proof.* The number of indices with $R(k) = q \geq 1$ is $(2q+1)^d - (2q-1)^d = O_d(q^{d-1})$. Thus the weighted sum is bounded by a constant plus a convergent series of order $q^{d-1-2r}$. For the compactness assertion, let $P_M$ retain the coordinates $R(k) \leq M$. On the radius-$B$ ball in $\mathcal{H}^{-r}$,

$$\|z - P_M z\|_{-s}^2 \leq (1+M)^{-2(s-r)} \|z\|_{-r}^2 \leq (1+M)^{-2(s-r)} B^2. \tag{10}$$

The projected ball is bounded in a finite-dimensional space and is therefore totally bounded. The tail tends uniformly to zero, so the full ball is totally bounded in the complete space $\mathcal{H}^{-s}$ and has compact closure there. □

For fixed finite $N$, each atomic lift belongs to $\mathcal{H}^{-r}$ when $r > d/2$: $|\widehat{X}_N(k)| \leq a^d \sum_x |\phi_N(x)|$ uniformly in $k$, and the sum of weights is finite. The map from the finite configuration to this carrier is a finite sum of carrier-valued Dirac coefficients, hence is continuous and Borel measurable. There is no need to define it by an arbitrary fallback value outside a set of finite configurations.

**Proposition 3.2** (Uniform tightness)**.** *For $d/2 < r < s$, the atomic laws are uniformly tight in $\mathcal{H}^{-s}$.*

*Proof.* Tonelli's theorem and Lemma 2.1 give

$$\sup_N \mathbb{E}\|X_N\|_{-r}^2 = \sup_N \sum_k w_r(k) \mathbb{E}|\widehat{X}_N(k)|^2 \leq L^d m^{-2} S_{d,r} =: C_{d,r,L,m} < \infty. \tag{11}$$

By Markov's inequality the probability of leaving the radius-$B$ stronger ball is at most $C_{d,r,L,m}/B^2$. Its weaker-space closure is compact by Lemma 3.1. Choosing $B$ large proves the claim. A bound with an extra factor two proves exactly the same tightness statement. □

3.1. **Why the approximation has a different regularity threshold.** An individual Dirac mass has coefficients of modulus one and lies in $\mathcal{H}^{-s}$ exactly when $s > d/2$. The same obstruction applies almost surely to a nonzero finite raw lattice lift. Indeed its Fourier coefficients are periodic modulo $N$. A nonzero configuration has a nonzero discrete Fourier coefficient, so on at least one residue class its coefficient has fixed positive modulus. The sum of $(1 + R(k))^{-2s}$ over that class diverges when $s \leq d/2$, for $s \geq 0$. The Gaussian configuration is nonzero almost surely, because its covariance is positive definite.

The continuum covariance decays as $\lambda(k)^{-1} \asymp |k|_2^{-2}$ at high frequency. Its expected squared $H^{-s}$ norm is therefore finite already for $s > d/2 - 1$. The limiting field can live on a smoother carrier than the atomic approximations. This improvement does not permit a weak-limit claim for the raw lifts on a space where those lifts are not random elements. Spectral truncation or interpolation changes the approximation and needs a separate theorem. Our formal choices deliberately lie on the safe atomic side: $(r, s) = (2, 3)$ in dimension two and $(3, 4)$ in dimension four.

## 4. Gaussian identification and full-sequence convergence

Choose one representative from each pair $\{k, -k\}$ of nonzero integer frequencies by requiring the first nonzero coordinate to be positive; call the resulting countable set $K_+$. Let $\xi_0$ and $(\xi_k, \eta_k)_{k \in K_+}$ be independent standard real Gaussians. Define

$$Z(0) = \frac{L^{d/2}}{m} \xi_0, \qquad Z(k) = \frac{L^{d/2}}{\sqrt{2\lambda(k)}} (\xi_k + i\eta_k), \qquad Z(-k) = \overline{Z(k)}. \tag{12}$$

There is one real zero coordinate, not a further independent zero sine coordinate. Tonelli and the mode variances give

$$\mathbb{E} \sum_k w_s(k) |Z(k)|^2 = L^d \sum_k \frac{w_s(k)}{\lambda(k)} < \infty \qquad (s > d/2).$$

The weighted series thus lies in the real carrier almost surely; finite partial sums converge there and define a measurable carrier-valued random element. Its law, denoted $\mu_{\mathrm{free}}$, is a Borel probability measure. The same summability argument, with the high-frequency mass symbol, gives finite expected norm for the broader range $s > d/2 - 1$ described above.

For a real smooth test, the pairing series converges in $L^2$; independence and conjugate symmetry show that it is centered real Gaussian. Its variance and polarized covariance are exactly (2). Equivalently, the characteristic functional is

$$\mathbb{E}e^{i\langle Z,f\rangle} = \exp\bigl(-Q(f,f)/2\bigr).$$

In particular, with $c_k(x) = \cos(2\pi k \cdot x/L)$ and $s_k(x) = \sin(2\pi k \cdot x/L)$ for $k \neq 0$,

$$Q(1,1) = L^d/m^2, \qquad Q(c_k,c_k) = Q(s_k,s_k) = L^d/(2\lambda(k)), \qquad Q(c_k,s_k) = 0.$$

These are physical pairings. They acquire an additional factor $w_s(k)$ if one takes variances of stored weighted coordinates instead.

**Lemma 4.1** (Finite Gaussian limits)**.** *For each finite collection of real and imaginary Fourier evaluations, the joint law of those evaluations of $X_N$ converges to that of $Z$.*

*Proof.* Both collections are centered real Gaussian vectors. The two complex second moments in Lemma 2.1 determine all real cross-covariances by polarization. For fixed labels the only eventual collisions are the genuine identities $k = \ell$ and $k = -\ell$. Moreover $\lambda_N(p_N(k)) \to \lambda(k)$ by Lemma 2.2. The covariance matrices therefore converge to those of (12). Their characteristic functions, $\exp(-t^{\mathsf{T}} C_N t/2)$, converge to the target Gaussian characteristic function, proving weak convergence in the finite-dimensional real space. □

**Lemma 4.2** (Fourier coordinates determine the law)**.** *Borel probability measures on $\mathcal{H}^{-s}$ are determined by all finite joint laws of the real coordinate at zero and the real and imaginary coordinates at labels in $K_+$.*

*Proof.* Those coordinates are continuous and reconstruct the negative labels by conjugation. They determine every finite projection in the real Hilbert space. The coordinate sigma-algebra equals its Borel sigma-algebra: coordinates are continuous in one direction; in the other, squared distances are countable sums of measurable coordinate functions, so balls are measurable. Separability makes all open sets countable unions of balls from a fixed countable base. Finite-dimensional cylinders form a generating intersection-stable family. The uniqueness theorem for probability measures on a generating family then proves the assertion. □

*Proof of Theorem 1.1.* Choose $d/2 < r < s$. The carrier is a separable complete real Hilbert space. Proposition 3.2 and Prokhorov's theorem give a weakly convergent subsequence inside every subsequence of the atomic laws; see, for example, Billingsley [3]. The continuous mapping theorem passes each finite coordinate projection to such a limit. Lemma 4.1 fixes those projected laws, and Lemma 4.2 identifies the weak limit as $\mu_{\mathrm{free}}$.

For completeness, uniqueness of subsequential limits alone is not the entire last step: tightness is also used. If the full sequence failed to converge, some neighbourhood of $\mu_{\mathrm{free}}$ would omit a subsequence. That subsequence has a further weakly convergent subsequence by tightness, whose limit must be $\mu_{\mathrm{free}}$, a contradiction. Hence the whole sequence converges. □

4.1. **Working on the full complex carrier.** The mathematical presentation took the closed real subspace as carrier. The four-dimensional Lean theorem is formulated on the full complex weighted $\ell^2$ completion, with the laws supported on its closed conjugate-symmetric subset. This requires an explicit support step. If $\mu_n(A) = 1$ for a closed set $A$ and $\mu_n \Rightarrow \mu$, Portmanteau applied to the open complement gives $\mu(A^c) \le \liminf_n \mu_n(A^c) = 0$. Canonical real coordinates are injective only on $A$; a measure comparison must therefore use this support information, not assert

injectivity on all complex sequences. The published proof establishes support passage, measurable coordinate encoding and the resulting real-supported measure extensionality before identifying arbitrary subsequential limits.

Both formal chains use the positive complex probe $C+iS$. Our negative exponential uses $C-iS$. The continuous map $z(k) \mapsto z(-k)$ converts the entire coordinate system, and the even symbol preserves both cross-moment identities. This is a joint coordinate transformation, not an inference of joint-law invariance from symmetry of separate sine marginals.

## 5. What is shared, and what changes in Lean

The common proof does not make the two native theorem statements identical. Table 1 records the relevant distinctions.

Table 1. The two verified instantiations. The weights refer to physical coefficients.

| Component | Two dimensions | Four dimensions |
|---|---|---|
| Physical site covariance | $a^{-2}G_N$ | $a^{-4}G_N$ |
| Raw lift | $a^2 \sum_x \phi_x \delta_{ax}$ | $a^4 \sum_x \phi_x \delta_{ax}$ |
| Moment / limit spaces | $H^{-2}$ / $H^{-3}$ | $H^{-3}$ / $H^{-4}$ |
| Native limit weight | 1 at zero; $R(k)^{-6}$ otherwise | $(1+R(k))^{-8}$ |
| Limit construction | Prokhorov existence plus identified cylinders | Named explicit Gaussian-series law |
| Identification | All finite weighted complex cylinders | Canonical real cylinders and real support |
| Sequence | $N = n+1$, $a = L/N$ | $N = n+1$, $a = L/N$ |

5.1. **The native two-dimensional theorem.** The final two-dimensional result asserts existence of a Borel probability measure on the concrete completion, convergence of the full lattice sequence to it, and equality of each finite Fourier cylinder with the corresponding periodic-resolvent Gaussian marginal. In the source, the main declaration is

`periodicHMinusThree_fixedVolume_weak_limit_exists_with_expected_cylinders`.

The finite cylinder reads the stored weighted complex coordinates. Its target is consequently obtained by complexifying cosine/sine Gaussian evaluations and multiplying each by the square root of the native weight. Dividing by that nonzero square root recovers physical Fourier marginals. The theorem has no caller-supplied measurability or tightness input.

Here the native stronger and weaker weights are

$$v_r(k) = \begin{cases} 1, & k = 0, \\ R(k)^{-2r}, & k \neq 0, \end{cases} \qquad r = 2, 3.$$

For integer frequencies, $w_r \leq v_r \leq 2^{2r} w_r$. Thus the native carrier is topologically equivalent to our common convention, but the norms are not literally equal. The full $H^{-2}$ energy includes the zero mode; a zero-mode-free seminorm would not control the stated Gaussian family.

5.2. **The native four-dimensional theorem.** The four-dimensional statement has the more explicit form

`fs4PeriodicFixedVolume_weak_tendsto_free`.

It proves convergence to the named law of a measurable Gaussian series on the full weighted completion. Positivity of $L$ and $m$ are its only explicit model hypotheses. The proof identifies every finite injective family of canonical real coordinates with actual periodic-resolvent Gaussian targets, passes real support to weak sublimits, and invokes real-supported cylinder uniqueness. The arbitrary-complex-carrier extensionality claim would be false for that reduced coordinate set and is not used.

The rank-four summability argument splits the integer lattice into two rank-two blocks. Mathlib's norm on a finite function type is a supremum norm: $\|(1,1)\|_\infty = 1$, not $\sqrt{2}$. The block expression in the weight therefore equals the coordinate maximum $R(k)$. This is compatible with a Euclidean covariance symbol because equivalent carrier norms do not change the topology. They must not be confused with equality of norms or rotation isometries.

5.3. **Reuse at the level of mathematical contracts.** Both developments use the same concrete finite-lattice covariance layer, physical test evaluations and periodic covariance limit. They then prove their own carrier membership, stronger-energy bound and compactness instance. The reusable contract is: construct actual probability laws on a Polish carrier, establish a stronger compactly embedded moment bound, identify all finite Gaussian evaluations, and prove those evaluations determine the law. The article abstracts this contract in Theorem 1.1; the current Lean library proves its two specified realizations. A generic Lean theorem for arbitrary dimension would require additional parametrized carrier and summability work.

## 6. Formal verification and reproducible artifacts

The source release contains two kinds of evidence. First, the Lean library is compiled with Lean 4.34.0-rc2 and Mathlib commit `852d20a84581aeba8527bd01d0be361d8c0e71c0`. Second, a portable release checker records the four exact four-dimensional declarations, their elaborated signatures, their axiom footprints, all source hashes, the dependency lock and the raw build and query logs. The checker has an offline mode and an isolated Lean replay mode. It rejects missing theorem names, changed signatures, custom axioms, `sorryAx`, source-hash drift and dependency revisions that do not match the lock file. The historical two-dimensional extension is checked against its preserved historical source bytes, rather than against the later expanded library root.

For the source snapshot cited in the bibliography, the module census is 94 modules, 18,475 module lines, 1,090 declarations, and 780 theorem/lemma declarations. The import-only library root is excluded from the line total. The source census contains no `sorry` and no project `axiom` declarations. These are reproducible engineering counts, not proof-strength metrics; the theorem contracts and their dependency footprints carry the mathematical evidence. The article's own `audit/bridge_contract.lean` additionally queries both native final theorems and type-checks examples of their convergence conclusions. Its executed transcript `audit/bridge_contract.log` and provenance record `audit/bridge_contract.json` bind that query to the release source hashes and exact dependency versions. Both final theorems have only the three foundational axioms listed below.

The final four-dimensional contract queries four declarations: the physical finite-marginal resolvent identity, probability-measure extensionality from finite real marginals, subsequential-limit identification, and the whole sequence theorem. The source transcript reports only the foundational axioms `propext`, `Classical.choice`, and `Quot.sound` for each. These are the ordinary foundational axioms used by Lean and Mathlib. The contract does not infer a mathematical theorem from a zero-`sorry` census: it checks the theorem names, types, dependencies and footprints explicitly.

The finite symbolic probe is complementary. It verifies character orthogonality, aliasing, the $L^d$ normalization, zero and Nyquist cases, the correct $L^{-4}$ dispersion factor, and finite rational instances of the shell estimates. It is a finite computation and cannot replace an all-$N$ proof. The proof guide and release metadata preserve this distinction.

TABLE 2. Lean components used by the comparison.

| Component | Role |
|---|---|
| `PeriodicHMinusThreeFixedVolumeTightness.lean` | Actual two-dimensional lattice laws, $H^{-2}$ moment control and tightness |
| `PeriodicHMinusThreeWeakLimitMarginal.lean` | Two-dimensional whole-sequence limit, Fourier cylinders and uniqueness |
| `PeriodicFS4DSobolevWeights.lean` | Four-dimensional max-coordinate weights and summability |
| `PeriodicFS4DCompactEmbedding.lean` | Compact inclusion from the stronger to weaker four-dimensional carrier |
| `PeriodicFS4DRealCarrier.lean` | Closed conjugate-symmetric carrier and compact real slice |
| `PeriodicFS4DProbeMoment.lean` | Actual lift, Gaussian target, real-coordinate encoding, marginals and final weak convergence |

6.1. **A reproducible article build.** The article itself is built by Tectonic from a seeded cache or an online bundle. The companion audit checks the source anchors, bibliography keys, compiled PDF page count, text extraction, unresolved references, and the presence of the main theorem and evidence-boundary phrases. It also recomputes the small-volume scale counterexample for the corrected dispersion estimate. The audit exits nonzero on any failed assertion.

```
python audit/audit.py --source-only
python build.py
python audit/audit.py --pdf build/main.pdf --repo ../..
```

*Evidence boundary.* The metadata and file names above are public release artifacts, not hidden assumptions about a reader's local AImath checkout. The article uses code names only where they identify a file or a fully qualified Lean declaration; the mathematical statements are written independently.

## Appendix A. Evidence ledger

This appendix records the provenance level of the article's principal claims.

Table 3. Claim-to-check ledger.

| Claim | Status | Independent check or boundary |
|---|---|---|
| Finite character orthogonality and $L^d$ factor | CONFIRMED | Symbolic probe and Lemma 2.1 |
| Correct physical-scale dispersion bound | CONFIRMED | Lemma 2.2; small-$L$ negative control |
| Uniform stronger-space moment bound | CONFIRMED | Paper Tonelli/Markov argument; Lean instantiations |
| Compact inclusion and tightness | CONFIRMED | Explicit tail estimate; dimension-specific Lean modules |
| Finite Gaussian covariance convergence | CONFIRMED | Exact Gaussian covariance calculation |
| Full arbitrary-$d$ theorem in Lean | UNVERIFIED | Paper synthesis only; no generic Lean declaration |
| Two-dimensional whole-sequence limit | CONFIRMED | Typed theorem contract and axiom transcript |
| Four-dimensional named-target whole-sequence limit | CONFIRMED | Typed theorem contract and axiom transcript |
| Exact all-mode complex identity in Lean | QUALIFIED | Paper identity; formal proof uses weaker real-probe bound |
| Infinite-volume or interacting four-dimensional limit | OUT OF SCOPE | No claim made |

The distinction between CONFIRMED and QUALIFIED is deliberate. A finite symbolic test can check a formula on rational instances but cannot prove its all-$N$ version. A Lean theorem can certify a fully elaborated declaration and its dependency footprint, but its scope is still the type that Lean checks. The article never promotes a source census or a successful PDF build into mathematical evidence.

The public release includes the exact source hashes, dependency lock, theorem transcript and build logs used for the Lean statements. Rebuilding the article is a separate artifact check. The ledger itself is therefore a guide to what each check establishes, not a replacement for the check.

## References


[1] S. Sheffield, *Gaussian free fields for mathematicians*, Probability Theory and Related Fields **139** (2007), 521–541. arXiv:math/0312099.

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